**Cambridge International Examinations**
Cambridge International General Certificate of Secondary Education

**CHEMISTRY** **0620/42**

Paper 4 Theory (Extended) **May/June 2016**

MARK SCHEME

Maximum Mark: 80

**Published**

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**Abbreviations used in the Mark Scheme**

- ; separates marking points
- / separates alternatives within a marking point
- **OR** gives alternative marking point
- **R** reject
- **I** ignore mark as if this material was not present
- **A** accept (a less than ideal answer which should be marked correct)
- **COND** indicates mark is conditional on previous marking point
- owtte or words to that effect (accept other ways of expressing the same idea)
- max indicates the maximum number of marks that can be awarded
- ecf credit a correct statement that follows a previous wrong response
- ( ) the word / phrase in brackets is not required, but sets the context
- ora or reverse argument

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| Question | Answer | Marks |
|---|---|---|
| 1(a)(i) | nitrogen / oxygen / fluorine / neon; | 1 |
| 1(a)(ii) | carbon; | 1 |
| 1(a)(iii) | oxygen; | 1 |
| 1(a)(iv) | nitrogen; | 1 |
| 1(a)(v) | neon; | 1 |
| 1(a)(vi) | carbon; | 1 |
| 1(a)(vii) | lithium / fluorine; | 1 |
| 1(a)(viii) | lithium; | 1 |
| 1(b)(i) | $B_2O_3$; | 1 |
| 1(b)(ii) | $Li_3N$; | 1 |

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| Question | Answer | Marks |
|---|---|---|
| 2(a)(i) | number of protons in one atom of an element; | 1 |
| 2(a)(ii) | **M1** number of protons and neutrons in one atom of an element;<br>**M2** in one atom of an element; | 2<br>1<br>1 |
| 2(b) | A: 6, 6, 6, $^{12}_{6}C$<br>B: 12, 12, 12, $^{24}_{12}Mg$;<br>C: 8, 10;, 8;, $^{16}_{8}O^{2-}$<br>D: 11, 10, 13, $^{24}_{11}Na^{+}$<br>11, 24; Na;+; | 6 |

| Question | Answer | Marks |
|---|---|---|
| 3(a)(i) | **M1** positive ions / cations (labelled or named in text);<br>**M2** electrons (labelled or named in text);<br>**M3** attraction between positive and negative; | 3<br>1<br>1<br>1 |
| 3(a)(ii) | (conduction due to) movement of electrons / mobile electrons; | 1 |
| 3(b) | $GaCl_3$;<br>$Ga_2(SO_4)_3$; | 2<br>1<br>1 |
| 3(c)(i) | $Ga_2O_3 + 6HNO_3 \rightarrow 2Ga(NO_3)_3 + 3H_2O$<br>formula of $Ga(NO_3)_3$; all formulae and balancing correct; | 2 |
| 3(c)(ii) | $Ga_2O_3 + 2NaOH \rightarrow Na_2Ga_2O_4 + H_2O$;<br>formula of $Na_2Ga_2O_4$; all formulae and balancing correct; | 2 |


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| Question | Answer | Marks |
|---|---|---|
| 3(d) | any 2 from:<br>• (do not) corrode;<br>• strong;<br>• hard;<br>• (improved) appearance; | 2 |

| Question | Answer | Marks |
|---|---|---|
| 4(a) | **M1** substance that speeds up a reaction / increases rate;<br>**M2** unchanged (chemically) at the end / not used up / lowers activation energy / provides alternative pathway; | 2<br>1<br>1 |
| 4(b) | **M1** too slow / slower;<br>**M2** lower yield / less product(s) / equilibrium shifts to left / equilibrium shifts in direction of reactants / backward reaction favoured / reverse reaction favoured; | 2<br>1<br>1 |
| 4(c) | faster / increase rate; | 1 |
| 4(d) | lower yield / less product(s) / equilibrium shifts to left / equilibrium shifts in direction of reactants / backward reaction favoured / reverse reaction favoured;<br>**OR**<br>higher cost / expensive;<br>**OR**<br>safety risks; | 1 |
| 4(e)(i) | **M1** breakdown of an ionic compound when molten or in aqueous solution;<br>**M2** (using) electricity / electric current / electrical energy; | 2<br>1<br>1 |
| 4(e)(ii) | carbon / graphite / platinum; | 1 |

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| Question | Answer | Marks |
|---|---|---|
| 4(e)(iii) | $2H^+ + 2e(^-) \rightarrow H_2$;<br>**OR**<br>$2H_3O^+ + 2e(^-) \rightarrow H_2 + 2H_2O$; | **1** |
| 4(e)(iv) | cathode / negative electrode; | **1** |
| 4(e)(v) | **M1** damp blue litmus paper;<br>**M2** bleaches / loses colour / turns white / turns colourless; | **2**<br>1<br>1 |
| 4(f) | $2NaCl + 2H_2O \rightarrow 2NaOH + H_2 + Cl_2$<br>all formulae correct; balancing; | **2** |
| 4(g) | **M1** *chlorine*:<br>treating (drinking) water / treating water in swimming pools / kill bacteria in water / chlorination of water / (manufacture of) paper products / plastics / PVC / dyes / textiles / medicines / antiseptics / insecticides / herbicides / fungicides / solvents / paints / disinfectant / bleach / hydrochloric acid;<br><br>**M2** *sodium hydroxide*:<br>drain cleaner / oven cleaner / extraction of aluminium / purification of bauxite / (manufacture of) biodiesel / paper / soap / detergents / washing powder / textiles / dyes;<br><br>**M3** *hydrogen*:<br>fuel / rocket fuel / fuel cells / in welding / (manufacture of) ammonia / $NH_3$ / margarine / methanol / hydrochloric acid / refrigerants; | **3**<br>1<br><br>1<br><br>1 |


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| Question | Answer | Marks |
|---|---|---|
| 5(a)(i) | more than enough to react (with all the hydrocarbon);<br>**OR**<br>(some) oxygen remaining; | **1** |
| 5(a)(ii) | 75 $cm^3$; | **1** |
| 5(a)(iii) | 2 : 15 : 10; | **1** |
| 5(a)(iv) | 2 : 15 : 10 : 10;<br>$C_5H_{10}$; | **2**<br>1<br>1 |
| 5(b)(i) | $C_7H_{16}$; | **1** |
| 5(b)(ii) | contains a double bond / triple bond / multiple bond;<br>**OR**<br>not all bonds are single bonds; | **1** |
| 5(b)(iii) | test: aqueous bromine / bromine (water) / $Br_2$;<br>result: (orange / yellow / brown) to colourless / decolourised / colour disappears; | **2**<br>1<br>1 |
| 5(c)(i) | addition; | **1** |
| 5(c)(ii) | 1 (kg); | **1** |
| 5(c)(iii) | propene: $CH_2$;<br>polypropene: $CH_2$; | **2**<br>1<br>1 |

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| Question | Answer | Marks |
|---|---|---|
| 6(a)(i) | roast/heat **and** in air/oxygen; | **1** |
| 6(a)(ii) | $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$;<br>$SO_2$ on right of equation; all formulae and balancing correct; | **2** |
| 6(b)(i) | **M1**<br>heat produced by carbon/coke (burning in) oxygen/air;<br>**OR**<br>$C + O_2 \rightarrow CO_2$ produces heat/exothermic;<br>**OR**<br>$2C + O_2 \rightarrow 2CO$ produces heat/exothermic (scores **M1** and **M2**);<br><br>**M2**<br>$C + CO_2 \rightarrow 2CO$;<br>**OR**<br>$2C + O_2 \rightarrow 2CO$;<br><br>**M3**<br>$ZnO + CO \rightarrow Zn + CO_2$;<br>**OR**<br>$ZnO + C \rightarrow Zn + CO$;<br>**OR**<br>$2ZnO + C \rightarrow 2Zn + CO_2$; | **3**<br>1<br><br>1<br><br>1 |
| 6(b)(ii) | temperature (inside the furnace) is above 907 °C/temperature (inside the furnace) is above the boiling point (of zinc)/1000 °C is above the boiling point (of zinc); | **1** |
| 6(b)(iii) | condensation/condensing/condense; | **1** |
| 6(c) | **M1** zinc is more reactive than iron/zinc is higher in the reactivity series than iron ora;<br>**M2** zinc loses electrons;<br>**M3** iron/steel/oxygen/air/water gains electrons **OR** electrons move to iron/steel/oxygen/air/water;<br>**M4** (therefore) iron does not lose electrons/get oxidised/form iron(II)/form iron(III); | **4**<br>1<br>1<br>1<br>1 |


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| Question | Answer | Marks |
|---|---|---|
| 6(d)(i) | green precipitate;<br>red-brown / brown / orange precipitate; | **2**<br>1<br>1 |
| 6(d)(ii) | oxidising agent / oxidant; | **1** |
| 6(d)(iii) | reducing agent / reductant; | **1** |
| 6(d)(iv) | iron(III) / $Fe^{3+}$; | **1** |
| 6(d)(v) | iron(II) / $Fe^{2+}$; | **1** |

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